DIALS: learning how to combine bundle evidence

Task: given some items of a bundle, rank the whole catalogue for the missing ones.

DIALS: content, label and co-occurrence terms, each learned, combined by a per-query gate; 41.6k parameters, one configuration on 8 datasets.

Questions this talk answers:

1. Combining history and descriptive evidence has been done before. What is different here?
2. Is it just an ensemble of existing scorers?
3. Does a 41.6k-parameter model have practical value against CLHE or RaMen?
4. Where does it not work?

From BundleFuse to DIALS

BundleFuse (presented last time): no trained scorer. Standardized evidence scores, weights chosen on validation, a frequency fallback $\frac{f_c}{f_c+\kappa}$. It beat published RaMen on POG and Spotify (+45% and +43% Recall@20).

What it showed: how evidence is combined matters more than model size; history and attributes are both needed.

BundleFuse, fixed by hand	What that could not do	DIALS, learned
One weight per source, per dataset	change the mix from query to query	a gate $w(S)$ per query
Fallback $\frac{f_c}{f_c+\kappa}$	adapt to how much history the seeds have	gate inputs plus a prior $\rho(f_c)$
Category evidence as similarity	learn "avoid the seed's category"	signed δ_l plus a label relation $P_l M_l Q_l^\top$
Recipe picked per dataset	one method for every dataset	one configuration on 8 datasets

DIALS keeps BundleFuse's evidence and learns its hand-set choices, with 41.6k parameters.

Why DIALS: what it offers

Bundle completion that still works when the items have no purchase history, at a fraction of the size of neural models.

- **Works without history.** On POG 96% of test queries have no co-occurrence evidence; there DIALS scores NDCG@20 .019 against CLHE's .005. On Spotify_sparse targets with no history: .263 against .003.
- **More accurate on identical splits and features.** Ahead of CLHE on all 6 datasets where it was rerun, and of RaMen on the 3 where it was: $+.014$ to $+.197$ NDCG@20, all $p = .002$.
- **Cheap to run.** 41.6k parameters (18–1,350$\times$ fewer), none per item; trains on CPU in 12–15 min on POG; a new item is scored on day one, without retraining.
- **Readable.** Every score splits into named terms and gate weights; the learned sign of the category term flips between domains (avoid the seed's category on fashion, prefer the same artist on music).

Where it stops: on warm grocery baskets (Dunnhumby) it ties the best simple baseline. The case for DIALS is new, sparse or changing catalogues.

The task and three kinds of evidence

$$S = \{\text{seed items}\} \longrightarrow \text{TopK}_{c \notin S}\ s(c \mid S)$$

Three kinds of evidence about a candidate c:

Evidence	Question it answers	Fails when
History (co-occurrence)	Was c bundled with these seeds?	c or the seeds are rare or new
Content (features)	Does c look like what goes with them?	resemblance $\neq$ compatibility
Labels (category, artist)	Does c's category go with theirs?	categories too coarse

The question is not whether to use all three, but how much to trust each one, for each query.

Is combining history and description new? No

Method	History	Labels	How the evidence is combined
Hybrid CF (content + ItemKNN)	co-occurrence	no	fixed weighted sum, one weight per dataset
CLHE (WSDM'24)	user–item LightGCN view	no	attention fuses ID, multimodal and CF views into **one item vector**
RaMen (ECAI'25)	item–item graph, hypergraph	no	fixed mixing constants, then **one item vector**
DIALS	bundle counts	yes	separate terms, **weights learned per query**, label relation learned

What is different is where and how the evidence is combined, not that it is.

What is different: separate terms, not one item vector

Early fusion (CLHE, RaMen)

$s(c \mid S) = \mathrm{enc}(S)^\top v_c, \quad v_c = \mathrm{fuse}(\mathrm{ID}, \mathrm{content}, \mathrm{graph})$

Evidence is mixed before scoring, so no query can say "these seeds have no history, trust content".

Late fusion with a gate (DIALS)

$s = w_C(S)\, C + \sum_l w_l(S)\, L_l + w_G(S)\, G$

Each evidence type keeps its own score, so seeds with no history still get content and label evidence.

Spotify_sparse, test NDCG@20 by share of targets with any co-occurring seed:

Share with history	0%	(0, 25%)	[25, 50%)	[50, 75%)	[75, 100%]
Queries	132	274	540	743	808
CLHE	.003	.022	.051	.104	.219
DIALS	**.263**	**.296**	**.218**	**.222**	**.298**
DIALS, co-occurrence switched off	.272	.305	.225	.213	.254
Gain from co-occurrence	−.009	−.009	−.007	+.009	**+.044**

Columns hold different queries: compare within a column. Co-occurrence switched off at inference, weights kept, 3 seeds; per-column differences not tested. The gate sees seeds, not which targets have history, so where few do, co-occurrence promotes wrong items.

What is different: labels, which CLHE and RaMen leave out

Simple label-aware rules (same label, label-to-label transition, blended with ItemKNN and content; tuned on validation)

Test NDCG@20 (clean)	POG_DENSE	POG	Spotify_sparse	Spotify
CLHE (rerun, same features)	.248	.0172	.115	.179†
Best label-aware simple rule	.249	.0234	.205	.247
DIALS	**.280**	**.0313**	**.256**	**.282**
Δ DIALS $-$ best rule	$+.031$	$+.008$	$+.051$	$+.036$
95% CI	[.028, .034]	[.006, .010]	[.046, .056]	[.031, .040]

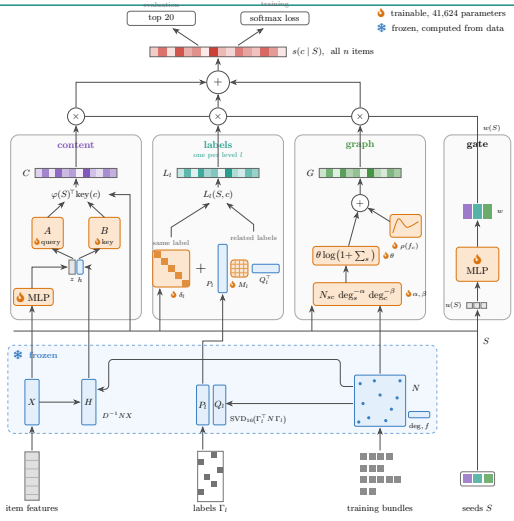
Holm over 8 tests (N and R): all $p = .0008$. †1 of 3 CLHE seeds finished on Spotify, all queries.

Labels alone beat CLHE; DIALS still beats the best rule that uses them.

Each design choice, and the evidence for it

Choice	Simpler alternative	Evidence
Separate learned terms	one fused item vector	no-history targets: CLHE .003, DIALS .263 (Spotify_sparse)
Per-query gate	fixed weights	$+$.016 NDCG on Spotify_sparse; $\leq$.006 elsewhere
Signed shared-label term δ_l	positive similarity	learned $\delta < 0$ on outfits, > 0 on playlists
Label factors fixed by SVD, only M_l learned	free label embeddings	free embeddings memorized the bundles of rare labels
Learned degree exponents α, β	raw counts	$\alpha, \beta \approx$ .3–.4 on POG, ≈ 0 on e-commerce
Leave-one-out, cross-fitting	train on own counts	without it a history term drove validation NDCG to .06
No second graph hop	two hops	level on 5 of 7 datasets, $-$.005 on Spotify; saves 2.5 min and 0.8 GB

DIALS: the architecture



Frozen (blue), from training data: features X, partner history $H = D^{-1}NX$, label factors P_l, Q_l (SVD of $\Gamma_l^\top N \Gamma_l$), counts N.

Trained (orange), three terms per item:

- content: maps A, B of $[z, h]$
- labels: δ_l plus $P_l M_l Q_l^\top$
- graph: damped counts plus prior $\rho(f_c)$

Gate: reads the seeds, outputs one weight per term.

Output: $s(c \mid S)$ for all n items; top 20, or softmax loss in training.

41,624 trained parameters; nothing per item.

DIALS: the score

$$
\begin{aligned}
s(c \mid S) = w_C(S)\, \underbrace{\phi(S)^\top \psi(c)}_{\text{content}} \\
+ \sum_l w_l(S) \Big[\underbrace{\delta_l(\cdot)\,[t_{g(c)}(S) > 0]}_{\text{shared label}} + \underbrace{\tfrac{t(S)^\top}{|S|} P_l M_l Q_l^\top e_{g(c)}}_{\text{label relation}} \Big] \\
+ w_G(S) \Big[\underbrace{\theta \log\left(1 + \textstyle\sum_{s \in S} n_{sc}\, d_s^{-\alpha} d_c^{-\beta}\right)}_{\text{co-occurrence}} + \underbrace{\rho(f_c)}_{\text{prior}} \Big]
\end{aligned}
$$

ϕ, ψ	rank-64 maps of item features and partner history
δ_l	small MLP; its **sign is learned** (substitute vs complement)
P_l, Q_l fixed, M_l (16×16) learned	which labels go with which
$w(S) = \mathrm{softplus}(\mathrm{MLP}(\cdot))$	per-query trust in each term

One configuration on all 8 datasets; no per-item parameters.

Training without leakage

1. **Leave-one-out.** A training bundle is scored with its own contribution removed from every count, degree, frequency and partner history.
2. **Cross-fitting.** The label SVD cannot be corrected that way, so a bundle being trained on uses factors computed without its fold (5 folds).
3. **Full-catalogue softmax.** Every hidden item competes with all items outside its bundle: no negative sampling.

Why it matters: an earlier version with free label embeddings memorized the bundles of rare labels; on Dunnhumby, a user-history term without leave-one-out drove training loss to 5.5 and validation NDCG to .06.

Results: against the best simple baseline

Dataset	Best simple (on validation)	Its N	DIALS N	ΔN [95% CI]	p (N)
Electronic	content+ItemKNN	.299	.335	+.036 [.016,.057]	.002
Food	content+ItemKNN	.425	.439	+.014 [−.006,.034]	.28
Clothing	content+ItemKNN	.330	.388	+.057 [.035,.080]	.002
Dunnhumby	content+ItemKNN	.104	.102	−.002 [−.005,.001]	.28
POG_DENSE	RP3β	.241	.280	+.040 [.036,.043]	.002
POG	EASE	.0198	.0313	+.012 [.009,.014]	.002
Spotify_sparse	content+ItemKNN	.153	.256	+.104 [.095,.112]	.002
Spotify	content+ItemKNN	.223	.282	+.060 [.054,.066]	.002

Clean test queries, 3 seeds, full-catalogue NDCG@20; paired cluster bootstrap, sign-flip test, Holm over 16 tests. Baselines: popularity, co-occurrence, ItemKNN, RP3β, GF-CF, EASE, iALS, SGFCF, blend.

Results: against CLHE and RaMen, rerun on identical splits and features

Test NDCG@20	CLHE	RaMen	**DIALS**	Δ vs best neural	Parameters (CLHE / RaMen / DIALS)
Electronic	.260	.254	**.335**	+.075	0.75M / 1.31M / 41.6k
Food	.272	.274	**.439**	+.164	0.78M / 1.38M / 41.6k
Clothing	.186	.191	**.388**	+.197	0.88M / 1.56M / 41.6k
POG_DENSE	.248		**.280**	+.033	7.25M / 11.6M / 41.6k
POG	.0172		**.0313**	+.014	9.48M / 16.1M / 41.6k
Spotify_sparse	.115		**.256**	+.141	29.0M / 56.5M / 42.0k

Their own code, 3 seeds each, settings chosen on validation; all 18 tests $p = .002$ (Holm over 20). Published numbers on the same POG/Spotify split files: RaMen POG .0226, Spotify .1882; DIALS (all queries) .0320, .2831. RaMen's e-commerce numbers (R@20 .84) use an unreleased split and a user–item graph that contains the test bundles.

With fewer training bundles, and with none

	POG_DENSE			POG		
Trained on	**DIALS**	CLHE	Best simple	**DIALS**	CLHE	Best simple
All bundles	**.280**	.248	.241	**.0313**	.0172	.0234
10% of bundles	**.239**	.202	.214	**.0113**	.0049	.0084
1% of bundles	**.201**	.169	.176	.0002	.0010	**.0016**
User histories, no bundles	.0046	.0028	**.0111**	.0013	.0020	**.0039**

- **Scarce bundles, dense data:** at 1% of POG_DENSE, DIALS (.201) is ahead of CLHE trained on the same 1% (.169), and not far below CLHE on all bundles (.248).
- **Too sparse:** at 1% of POG every method is near zero; DIALS is slightly behind.
- **No bundles at all:** trained on user histories, every method is near zero. Histories pair *substitutes* (21–26% of pairs share a category), bundles pair *complements* (≈0%).

Clean test NDCG@20, 3 seeds; subsets nested, same validation and test queries. DIALS vs CLHE and vs best simple: $p \leq .017$ (Holm over 40) except: POG histories vs CLHE ($p = .38$); with users who overlap test bundles removed, DIALS vs CLHE not separated on either dataset ($p \geq .37$). Best simple chosen on validation per condition.

Is it just an ensemble? The final step is a sum; the test is what is summed

Test NDCG@20 (clean), 3 seeds

	POG_DENSE	POG	Spotify_sparse
Standard scorers, tuned weighted sum[c]	.249	.0234	.205
DIALS terms, **fixed** weights	.277	.0308	.240
DIALS terms, **per-query gate**	**.280**	**.0313**	**.256**

- **Same evidence, standard scorers:** the best tuned weighted sum loses .008–.051 to DIALS ($p = .0008$ on all 4 datasets).
- **Same learned terms, fixed weights:** keeps most of the gain. The gain is in the learned terms.
- **The gate adds** +.016 on Spotify_sparse, +.003 on POG_DENSE, +.002 on Dunnhumby (Holm $p \leq .0014$); not separated on Electronic, Food, Clothing, POG.

[c] Label rules + ItemKNN + content, weights tuned on validation. Fixed weights = each dataset's mean gate on validation queries; trained model otherwise unchanged; Holm over 14 tests.

What a fixed ensemble cannot do: the relation changes sign by domain

Dataset	δ: one of 3 seeds shares c's label	Gate weights: content / labels / graph
POG	-0.16	4.2 / **14.8** / 3.7
POG_DENSE	-0.31	6.0 / **15.2** / 3.1
Food	$+1.63$	1.2 / 1.1 / **4.7**
Clothing	$+1.91$	1.9 / 1.6 / **6.6**
Spotify (artist)	$+2.63$	**3.1** / 1.0 / 1.1

- **Outfits:** a shared category is evidence *against* (one item per category). Nothing in the model says this.
- **Playlists and e-commerce bundles:** a shared artist or category is evidence *for*.
- An ensemble of similarity scorers can only add "same category" with a positive weight; DIALS learns the sign, with one configuration on every dataset.

Why it works: where the lead comes from

Test NDCG@20 by the most co-occurrence any seed has with the target

Dataset		0	1	2–3	4+
POG_DENSE	share of queries	36%	15%	14%	35%
	CLHE / **DIALS**	.137 / **.151**	.249 / **.265**	.302 / **.328**	.339 / **.400**
POG	share of queries	96%	3%	1%	0.3%
	CLHE / **DIALS**	.005 / **.019**	.271 / .261	.458 / .556	.578 / .666

- **POG is a no-history problem:** 96% of queries have no co-occurrence evidence; there DIALS is 4× CLHE.
- **The gate responds:** on Spotify_sparse the graph-to-content weight falls from 0.43 to 0.16 as the share of unseen seeds rises.

What each term contributes (inference-time knockouts)

Test NDCG@20 (clean)	POG_DENSE	POG	Spotify_sparse
Full DIALS	**.280**	**.0313**	**.256**
without content	.239	.0198	.232
without labels	.261	.0203	.137
without label low-rank only	.274	.0269	.256
without co-occurrence	.263	.0239	.242
without prior	.051	.0161	.238

Every term matters somewhere, and which one matters most depends on the domain: the prior on POG_DENSE, labels on Spotify, content and labels together on POG.

Trained weights kept; nothing retrained. 3 seeds.

Practical value: what DIALS costs to run

	DIALS	CLHE	RaMen
Trained parameters (POG)	**41.6k**	9.48M	16.1M
Per-item parameters	**none**	2 tables $n \times 64$	256 per item
Training on POG (4 CPU threads)	**12–15 min**	85 min	71 min*
Inference, ms/query on CPU (POG / Spotify)	0.49 / 3.24	0.53 / –	0.69 / –
Full Spotify training	CPU, hours	≈6.5 GPU-h per seed	–

- **New items on day one:** scored from features and category, no retraining.
- **Explainable:** each recommendation decomposes into named terms and gate weights.
- **Honest cost:** memory is dominated by the co-occurrence matrix: about 560 MB on Spotify (about 260 MB compressed), against about 130 MB for CLHE and 260 MB for RaMen.

*RaMen: 21 s per epoch measured, times its configured 200 epochs; not trained to completion by us.

Where DIALS does not win

- **Dunnhumby (grocery baskets):** ties the content+ItemKNN blend ($-.002$ NDCG, $p = .28$). Warm and co-occurrence-driven; 97% of items are in training baskets.
- **Personalization helps everyone equally:** adding the household's repurchase counts lifts DIALS by 45% ($.102 \rightarrow .148$), and the blend by 42% ($.104 \rightarrow .148$): a tie again.
- **Food:** recall separated, NDCG not ($p = .28$).
- **Features matter on POG:** the released BLIP text features on Chinese titles cost DIALS 13% against a Chinese text encoder (.0313 vs .0360).
- **No bundles, no model:** trained on user histories instead of bundles, every method is near zero; simple RP3β or label rules come out ahead of DIALS.
- **Open:** CLHE on full Spotify has 1 of 3 seeds; RaMen not rerun on POG/Spotify.

Contribution, stated so it can be defended

1. **Separate, jointly learned evidence terms** (late fusion) instead of one fused item vector; a per-query gate on top, which matters where history varies across queries (+6% on Spotify_sparse).
2. **Labels as a learned relation:** a signed diagonal (substitute vs complement) plus a low-rank label-to-label map on cross-fitted spectral factors.
3. **One configuration** that recovers opposite domain structure (avoid the seed's category on fashion, same artist on music) on 8 datasets.
4. **Leakage-free training** by leave-one-out and cross-fitting.

The components exist separately (graph filters with degree exponents, category compatibility, homophily/heterophily learning); the claim is the combination and its measured effect, not that any part is first.

In one line: bundle completion that works without purchase history, with 41.6k parameters.

Next: Instacart (real product names and aisles); TIFU-KNN and UP-CF@r for the personalized task; RaMen on POG and Spotify.

Backup: other published protocols

Bundle-MLLM (KDD'25), HitRate@1 among 10 candidates, released split files:

	DIALS	Bundle-MLLM (published)	CLHE (in that paper)
POG	**.630**	.413	.348
POG_DENSE	**.781**	.668	.637

DDBC (ICLR'26), F1 within a candidate pool:

	DIALS	Best simple	DDBC (published)
POG, $k = 4$	**.488**	.452	.139
Spotify, $k = 30$	**.625**	.540	.282

Published numbers, not rerun. Bundle-MLLM states an 8:1:1 split; only 7:1:2 files are released, and those are used.